

Kyle Shiroma

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Education

University of California, San Diego

B.S. in Data Science, Minor in Mathematics

La Jolla, CA

Expected June 2027

Experience

HDSI Lab 3.0, University of California, San Diego

Data Science Fellow

San Diego, CA

October 2025 – Present

- Engineering a computer vision pipeline to detect and **auto-correct misalignment** in **400+ LED arrays** using laser sensor feedback and classification models.
- Designed and built an **interactive poster board demo** using 3D-printed linear actuators, servo motors, and fiber-optic pathways (LED) to visualize a 3D optical sensing system.

CIC-Summer Research Program, California State University, Fullerton

Data Science Research Intern

Fullerton, CA

May 2024 – July 2024

- Built and evaluated **machine learning models (Logistic Regression, XGBoost, Random Forest)** on **5,900+ UEFA matches** using **engineered features** (dynamic Elo ratings, venue indicators).
- Achieved **61% accuracy** with **20% recall improvement** for the rare "Draw" class; identified **Elo difference** as the most influential feature through **feature importance analysis**.

Projects

Customer Segmentation with RFM Analysis

- Analyzed **500,000+ retail transactions** across **37 countries** using SQL-based ETL transformations, Excel exploratory analysis, and RFM segmentation to identify **five distinct customer behavioral segments** (Big Spenders, At Risk, Recent One-Time Buyers).
- Translated raw sales data into executive-ready **Tableau** dashboards, enabling marketing teams to optimize spend allocation by 25% and identify at-risk customer segments for targeted retention..

Pulsepanion – AI4Purpose NYC Hackathon Winner (Team Lead)

- Led **cross-functional team** in developing bilingual R Shiny dashboard enabling **caregivers** to upload patient health data and generate clear, actionable summaries in **less than 15 seconds**.
- Integrated Python scripting and OpenAI-LLM powered answers via reticulate, **reducing manual review time by 70%** and improving caregiver workflow efficiency.
- **Presented solution to industry judges and healthcare stakeholders**, clearly communicating technical approach, user benefits, and business value.

Skills

Technical Skills: Python, SQL, R, JavaScript; Feature Engineering, Model Evaluation, A/B Testing

Data Visualization & Communication: Tableau, Excel, Google Sheets

Tools & Platforms: Git, PostgreSQL, SQLite, Microsoft 365 (Excel, PowerPoint, MS Project)

Certifications

- **Apache Airflow: The Hands-On Guide (Udemy)** – Building scalable data pipelines using DAGs, Spark, Docker, and AWS S3 with ETL process.